Spokes & Flows: Milestone 2, Data Analysis and Sketches

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Code: github.com/HarshithKoriRaj/spokesnflows (private, shared with `ctsilva` and `kimryan0416`)

1 What changed since Milestone 1

The feedback said our plan was too big for two people and that our best line was "where does the system struggle, and when?" We used that sentence to decide what to keep. Every change is also marked with a yellow *Changed since M1* box next to the question it affects.

Audience. Our main reader is someone planning the Citi Bike network, such as NYC DOT staff or a community board deciding where to add docks. They need to see which stations keep losing or gaining bikes through riders' trips, at what hours, and where those trips come from. A second reader is a regular rider who wants to know which stations tend to drain at their commute time.

Questions we kept, merged, refined and dropped.

- **Q1. When do people ride?** Kept. It sets up the daily rhythm the other views rely on.
- **Q2. Where and when do stations gain or lose bikes through trips?** Our old Q2 (where trips happen) and Q3 (imbalance) are now one map, as the feedback suggested.
- **Q3. Where do the trips behind that imbalance come from?** Our old Q4. We planned to focus on East River crossings, but the data showed they are only about 6.5% of weekday morning trips (Table 6). Most trips go to a nearby neighborhood in the same borough. So the question is now about neighborhood-to-neighborhood flows in general, and East River crossings are just one filter.
- **Q4. How much do weather and season change ridership?** Our old Q6. Kept because it is cheap (one row per day) and it tells a planner how much of a busy or quiet day is just the weather.
- **Q5 (optional). Do e-bike trips differ from classic-bike trips?** Our old Q5. The data part is done because it comes from the same cleaned table. We will only build the view if time allows.
- **Dropped:** our old Q7 (whether new stations connect to the rest of the network). It needs a second year of data and a network layout, and it doesn't help answer the main question. We also did not take on the optional congestion-pricing extension.

Three months instead of a full year. We use January, July and October 2025: one winter, one summer and one fall month, 11,844,428 trips in total. This covers the seasonal range for Q4 while keeping the processing light. Our pipeline works on any set of months, so the code would not change if we added more.

2 Data handling

2.1 Sources

- Citi Bike trip histories published by Lyft. Monthly files from the public S3 bucket: `202501`, `202507` and `202510-citibike-tripdata.zip`. One row is one trip.
- Dock capacity from the Citi Bike GBFS feed (`station_information`, downloaded November 2026).
- 2020 Neighborhood Tabulation Areas from NYC Open Data (262 areas, with borough names).
- NOAA daily weather, Central Park: daily high and low temperature, rain and snow.

How to get each file is listed in the repository README.

2.2 Pipeline

Everything runs in two scripts. `scripts/download.py` fetches the files. `scripts/build.py` converts each month to Parquet, loads it into DuckDB, cleans it, joins the other datasets and writes the small tables the visualizations will read. It also writes every table and number in this report. A full run on the three months takes about 40 seconds on a laptop.

1. **Filter:** keep trips that start in one of the three months (each file also holds a few trips that began the evening before), and drop trips longer than 3 hours, which are almost always bikes that weren't docked properly. The operator already removes trips under 60 seconds.
2. **Derive:** trip duration, straight-line distance, start and end hour, day of week, and weekday/weekend.
3. **Stations:** key each station on its ID, take its most common name, and use the median coordinates of classic-bike trips (classic bikes report the dock's position; e-bikes report their own GPS).
4. **Join:** dock capacity from GBFS (by station ID, then by location within 75 m), neighborhood and borough by point-in-polygon, and weather by date.
5. **Aggregate:** write the processed tables in Table 1.

Processed table	One row is	Rows	Used by
`hourly_profile.csv`	day of week × hour × rider type	336	Q1
`station_hour.csv`	station × weekday/weekend × hour	114,144	Q2
`stations.csv`	station (location, docks, neighborhood)	2,378	Q2, Q3
`od_nta.csv`	neighborhood pair × time period	32,054	Q3
`daily.csv`	day, with weather	93	Q4
`bike_type_summary.csv`	bike type × rider type	4	Q5
`distance_hist.csv`	bike type × 0.25 km distance bin	66	Q5

Table 1: Every view reads from one of these tables, and all of them come from the same cleaned trip table.

2.3 What we found when checking the data

Check	Trips	% of raw	What we do
Trips in the downloaded files	11,844,428	100.00	–
Started outside the downloaded months	1,626	0.01	dropped
Duplicate ride IDs	0	0.00	dropped (kept first)
Shorter than 1 min	0	0.00	dropped
Longer than 180 min	16,130	0.14	dropped
No start station ID	6,704	0.06	kept; not in station views
No end station ID	37,704	0.32	kept; not in station views
of which e-bikes	35,368	0.30	
of which classic bikes	2,336	0.02	
with end coordinates	38	0.00	
Round trips (same start and end)	228,940	1.93	kept, flagged
Trips after cleaning	11,826,946	99.85	–
Stations found in trips	2,378		
capacity matched by ID	2,185		
capacity matched by location	123		within 75 m
capacity missing	70		shown as "no data"
outside any NYC neighborhood	68		

Table 2: Each check, how many trips it affects, and what we do. Percentages are of the raw trip count.

Trips with no end station (feedback point 3). 37,704 trips (0.32%) have no end station, and 94% of them are e-bikes. We expected these to be e-bikes left outside a dock, but 37,666 of them also have no end coordinates, so we can't tell where they ended. Only 38 trips end at a known location away from a station. The classic-bike trips in this group last about 25 hours, which looks like the system closing trips for bikes that were never returned. So the 2025 files have essentially no "dockless" trips with a known end point. We keep these trips in the time and weather views (Q1, Q4) and leave them out of the station and neighborhood views (Q2, Q3), which slightly undercounts arrivals there. Likewise, 6,704 trips have no start station and no start coordinates. They are handled the same way.

Stations outside NYC. 68 stations fall outside every NYC neighborhood. They are Jersey City and Hoboken docks that some NYC trips ended at. They stay out of the map, and those trips count as leaving NYC.

Dock capacity. The GBFS feed lists today's stations, not 2025's. 67 NYC stations still have no capacity after matching. 26 of them are listed with zero docks (probably closed or temporary), and 41 are no longer in the feed. A few are busy, such as West St & Chambers St. These stations stay on the map in grey, labeled "no dock data", and get no color.

2.4 What our imbalance measure shows, and what it doesn't (feedback point 2)

For each station and hour we compute **net trip flow** = arrivals − departures, averaged over the 69 weekdays (or 24 weekend days), and divide by the number of docks. Adding it up from midnight gives the **cumulative net trips per dock** at each hour. A value of -0.8 at 10:00 means that, on an average weekday, riders took 0.8 more bikes per dock out of the station than they brought back since midnight.

This measures *trips*, not how many bikes were at the station. Our own data shows the gap clearly. By 10:00 on weekdays, North Moore St & Greenwich St in Tribeca gains 3.65 bikes per dock through trips (Table 5). That is far more bikes than the station can hold, so staff must be taking bikes away (by rebalancing or valet service). Rebalancing trips are not in the public data. So we will say a station "loses (or gains) bikes through riders' trips", never that it is "empty" or "full", and the map will carry a note saying so.

3 Domain questions: processed data and sketches

For each question we give the unit of observation, the transformation, a sample of the processed table, and our sketch. Across all views: panels that are compared share one scale, missing data is drawn as its own grey category, and every color difference is also shown another way (shape, line style or position).

Q1. When do people ride, and how do members and casual riders differ?

Unit: day of week × hour × rider type. **Transformation:** count trips by start hour and weekday, then divide by how many times that weekday appears in the three months to get trips per day. **What the sample shows:** on weekdays members have two peaks, about 9,200 trips in the 8:00 hour and 11,000 at 17:00. Casual riders barely have a morning peak (about 900 at 8:00) but rise in the evening. **View:** two heatmaps side by side (members, casual riders), hour across and day of week down. Color is each hour's share of that group's weekly trips, so the shapes can be compared even though members take 83% of all trips. Panel titles give trips per day. Hovering a cell shows the exact count.

day	hour	rider type	trips (study period)	trips per day
Wednesday	8	casual	13,795	920
Wednesday	8	member	146,149	9,743
Wednesday	17	casual	28,786	1,919
Wednesday	17	member	183,567	12,238
Saturday	8	casual	7,544	629
Saturday	8	member	35,775	2,981
Saturday	17	casual	28,020	2,335
Saturday	17	member	80,107	6,676

Table 3: Sample of `hourly_profile.csv` (Wednesday and Saturday at 8:00 and 17:00). The full table has 336 rows.

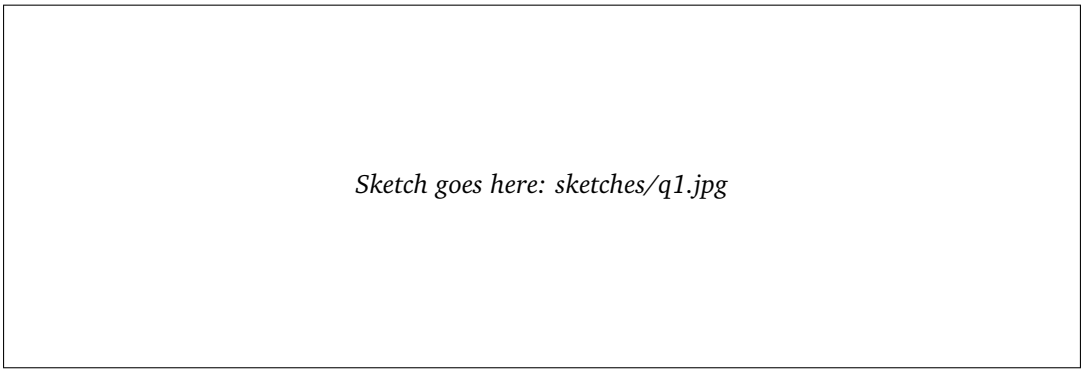


Figure 1: Q1 sketch: hour × weekday heatmaps for members and casual riders.

Q2. Where and when do stations gain or lose bikes through trips?

Changed since M1: merges old Q2 and Q3 into one map; the measure is now explicitly a trip measure, normalized by docks.

Unit: station × weekday/weekend × hour (2,378 stations × 48). **Transformation:** departures are counted at the start station and start hour, and arrivals at the end station and end hour. Both are divided by the number of days. Net and cumulative net are then divided by the number of docks. Every station has all 24 hours, so hours with no trips appear as zero instead of disappearing. **What the sample shows:** the busiest station, W 21 St & 6 Ave (74 docks), gains about 0.66 bikes per dock through trips by noon on weekdays. Across all stations with dock data, 91 lose more than half a bike per dock by 10:00 and 79 gain more than that. The top losers are in residential areas such as the Upper East Side and Greenpoint. **View:** a map of stations colored on a diverging scale centered at zero (red = losing, blue = gaining, legend in net trips per dock). Losing stations are drawn as downward triangles and gaining ones as upward triangles, so direction doesn't depend on color. Stations with no dock data are grey hollow circles. An hour slider and a weekday/weekend toggle sit above the map. Clicking a station opens a bar chart of its departures and arrivals by hour, with a line for cumulative net per dock.

station	hour	dep/day	arr/day	net/day	docks	net/dock	cum. net/dock
W 21 St & 6 Ave	6	9.9	15.5	5.6	74	0.08	0.16
W 21 St & 6 Ave	7	16.6	19.9	3.3	74	0.04	0.20
W 21 St & 6 Ave	8	34.1	46.9	12.8	74	0.17	0.37
W 21 St & 6 Ave	9	31.2	45.1	13.9	74	0.19	0.56
W 21 St & 6 Ave	10	19.7	24.5	4.8	74	0.06	0.63
W 21 St & 6 Ave	11	19.9	22.1	2.2	74	0.03	0.66

Table 4: Sample of `station_hour.csv`: the busiest station, weekday mornings (per average weekday).

station	borough	docks	cum. net/dock by 10:00
3 Ave & E 82 St	Manhattan	43	-0.96
Franklin St & Dupont St	Brooklyn	27	-0.82
E 20 St & FDR Dr	Manhattan	63	-0.81
W 50 St & 9 Ave	Manhattan	18	-0.80
North Moore St & Greenwich St	Manhattan	29	3.65
Dock 72 Way & Market St	Brooklyn	22	2.75
Greenwich St & Hubert St	Manhattan	35	2.29
E 47 St & Park Ave	Manhattan	81	1.55

Table 5: Stations with at least 15 docks that lose or gain the most bikes per dock through trips by 10:00 on an average weekday.

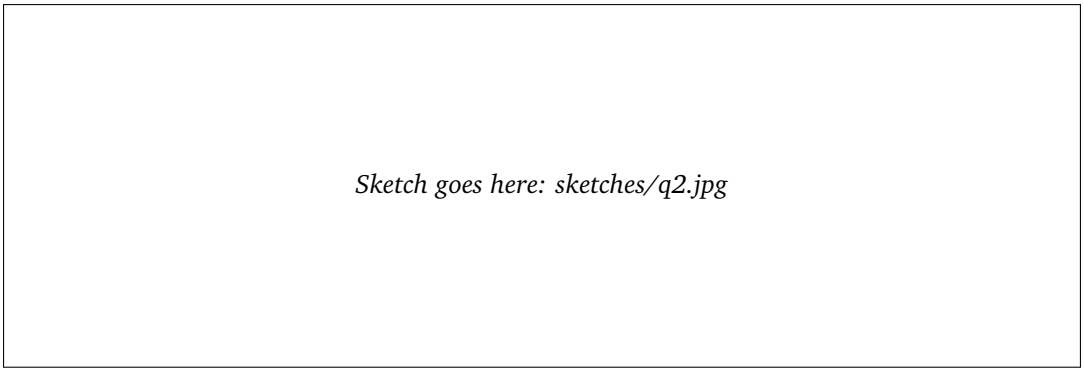


Figure 2: Q2 sketch: net-flow map with hour slider and weekday/weekend toggle, and the hourly profile shown when a station is selected.

Q3. Where do the trips behind that imbalance come from?

Changed since M1: we dropped the East River crossings as the focus because they are only about 6.5% of weekday morning trips (Table 6). Crossings are now a filter, and the question covers all neighborhood-to-neighborhood flows.

Unit: start neighborhood × end neighborhood × time period (weekday 7–10, weekday 16–19, other weekday hours, weekend). **Transformation:** join each trip's start and end station to its 2020 neighborhood and borough, count trips per pair and period, divide by the number of days, and flag pairs that cross a borough line or the East River. **What the sample shows:** about 70% of trips go to a different neighborhood in the same borough and 21% stay in the same neighborhood. The busiest morning flows are short hops into Midtown, from Chelsea-Hudson Yards, Hell's Kitchen and Murray Hill. On weekday mornings Brooklyn sends about 1,040 trips per day into Manhattan, against about 15,200 trips within Manhattan. **View:** a borough-to-borough matrix on the left and a map of the busiest neighborhood flows on the right. Flows

are curved lines whose width is trips per day. East River crossings are solid and other lines dashed. A period selector changes both views, and clicking a neighborhood keeps only its flows. Selecting a station in Q2 highlights its neighborhood here.

period	trips/day	same nbhd %	same borough %	other borough %	East River %
Weekday 7–10	24,046	21.3	69.7	9.0	6.5
Weekday 16–19	33,997	21.2	69.8	9.0	6.5
Weekend (all day)	119,712	21.7	68.5	9.9	6.9

Table 6: Where trips go, by period (trips with a known start and end neighborhood). "Same borough" means a different neighborhood in the same borough.

from neighborhood	to neighborhood	trips/day	East River
Chelsea-Hudson Yards	Midtown South-Flatiron-Union Square	228.6	no
Chelsea-Hudson Yards	Midtown-Times Square	219.7	no
Hell's Kitchen	Midtown-Times Square	216.8	no
Greenpoint	Williamsburg	193.9	no
East Village	Greenwich Village	186.1	no
Financial District-Battery Park City	Tribeca-Civic Center	172.8	no
Murray Hill-Kips Bay	Midtown-Times Square	166.2	no
West Village	Chelsea-Hudson Yards	160.9	no

Table 7: Sample of `od_nta.csv`: the busiest pairs of different neighborhoods, weekday 7–10.

from / to	Bronx	Brooklyn	Manhattan	Queens
Bronx	464	0	107	1
Brooklyn	0	5,161	1,042	216
Manhattan	104	223	15,157	59
Queens	2	167	232	1,109

Table 8: Weekday 7–10 trips per day between boroughs (rows: start, columns: end).

Sketch goes here: sketches/q3.jpg

Figure 3: Q3 sketch: borough matrix and neighborhood flow map with period selector.

Q4. How much do weather and season change ridership?

Changed since M1: uses three months (January, July, October) instead of the full year.

Unit: one day (93 days). **Transformation:** count trips per day by rider type and bike type, then join that day's high temperature, rain and snow from Central Park by date. Holidays are flagged so they aren't mistaken for weather effects. **What the data shows:** an average day had about 68,500 trips in January, 160,600 in July and 152,500 in October. On rainy weekdays (0.1 inch or more), members rode at 88% of their usual level for that month, but casual riders dropped to 68%. **View:** a calendar heatmap of daily trips for the three months (weeks across, days of the week down, holidays outlined). Below it, two scatter plots share one y-axis: members and casual riders, each showing trips as a percent of that group's monthly average against the day's high temperature. Rainy days are filled dots and dry days hollow.

date	day	trips	members	casual	high (F)	rain (in)	snow (in)
2025-01-01	Wednesday	51,388	40,516	10,872	51	0.00	0.00
2025-01-02	Thursday	71,741	63,312	8,429	43	0.00	0.00
2025-01-03	Friday	75,319	66,316	9,003	39	0.00	0.00
2025-01-04	Saturday	55,363	48,153	7,210	33	0.00	0.00
2025-01-05	Sunday	52,155	45,968	6,187	33	0.00	0.00
2025-01-06	Monday	58,491	54,448	4,043	30	0.08	0.90
2025-01-07	Tuesday	65,183	60,766	4,417	33	0.00	0.00
2025-01-08	Wednesday	66,561	62,083	4,478	30	0.00	0.00
2025-01-09	Thursday	67,342	62,566	4,776	33	0.00	0.00
2025-01-10	Friday	81,345	73,504	7,841	39	0.00	0.00

Table 9: Sample of `daily.csv` (first ten days of January; Jan 1 is a holiday).

Sketch goes here: sketches/q4.jpg

Figure 4: Q4 sketch: calendar heatmap and trips vs. temperature for members and casual riders.

Q5 (optional). Do e-bike trips differ from classic-bike trips?

Unit: bike type $\times$ rider type (summary), and bike type $\times$ 0.25 km distance bin (histogram). **Transformation:** straight-line distance between start and end coordinates, which is shorter than the real route. **What the data shows:** e-bikes carry about 71% of trips. For members, the median e-bike trip covers 1.63 km against 1.06 km on a classic bike, and takes only about one minute longer. **View:** overlaid step histograms of trip distance, solid for e-bikes and dashed for classic bikes, with the medians marked.

bike	rider	trips	median min	median km	90th pct km	no end station
classic	casual	489,347	14.70	1.52	3.81	0.00
classic	member	2,985,142	7.47	1.06	3.13	0.00
electric	casual	1,555,785	11.72	1.77	4.55	0.01
electric	member	6,796,672	8.45	1.63	4.42	0.00

Table 10: `bike_type_summary.csv`. "No end station" is the share of trips without an end station.

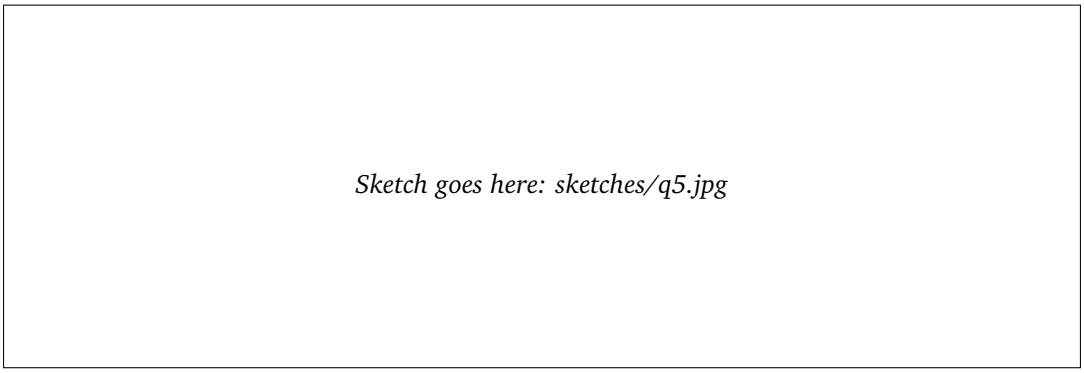


Figure 5: Q5 sketch (optional): trip distance by bike type.

AI use

We used Claude (Anthropic) to help write the two data scripts in our repository's `scripts/` folder (`download.py` and `build.py`), to run the first data checks, and to draft parts of this report's text. We reviewed and ran the code, checked the outputs against the raw files, and edited the text, and we can explain every step. The sketches are our own drawings, made without AI. The same statement is in the repository's README.